

BRIAN YUAN-AN LIU

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GEN AI & CYBERSECURITY SPECIALIST

Strategic technologist with 10+ years of experience delivering scalable cloud architectures and enterprise-grade Generative AI solutions. Combines deep Senior Staff Software Engineering leadership (Zscaler, AWS, F5) with dual expertise in AI innovation and cybersecurity to align business objectives with high-impact technical execution. Expert in designing production-ready LLM workflows (RAG, agentic systems, vLLM/SGLang execution engines), optimizing high-throughput cloud infrastructure, and embedding zero-trust and kernel-level security protections. Proven track record driving the evolution from manual engineering practices to automated, agentic workflows and leading high-level system architecture that drives business growth, reliability, and security resilience.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY, Chicago, IL

Master of Science, Computer Science

Aug 2016 - May 2018

Association: IIT-Security Lab

CHUNG YUAN CHRISTIAN UNIVERSITY, Taoyuan, Taiwan

Bachelor of Science, Computer Science

Sep 2011 - Jun 2015

Association: CYCU-Software Engineering Lab

SKILLS

- **Languages:** Go, Python, C++, C, Shell Script, TypeScript, Java
- **Cloud & Infrastructure:** AWS, Azure, GCP, CDK, Terraform (IaC), Kubernetes, Docker, Helm, Nginx/Traefik, Bedrock, SDN, OVS, eBPF, GNU toolchain, BMaaS, Distributed/Embedded Systems
- **AI/ML/Inference:** Gen AI, LLM, ML, RAG, NLP, RAGAS, Agentic AI, LangChain, LangGraph, Hugging Face, GPT, Claude, Gemini, LLaMA, TensorFlow, PyTorch, Fine-Tuning(LoRA/ QLoRA), SGLang, vLLM, Spec Driven Development (SDD), Local/Edge LLM, Quantization (AWQ/ GGUF)
- **Networking/Protocols/Security:** 5G, gRPC, NSH, SFC, VLAN, VxLAN, TCP/IP, REST, SOAP, P2P, ZMQ, SRv6, Zero Trust (mTLS/TLS), CTEM, SIEM, OWASP, SQL injection, prompt injection, header injection, Kernel & Application Security (memfd_create, XSS, CSP and more)
- **Data & Services:** DynamoDB, PostgreSQL, MySQL, PostgresSQL, OpenSearch/ElasticSearch, Git, Kafka, Kinesis, Protobuf, vim, Telemetry, Prometheus, Temporal, Atlas

EXPERIENCE

ZSCALER, INC. San Jose (Remote), CA

Nov 2024 - present

Senior Staff Software Development Engineer (2026 - present)

Senior Software Development Engineer (2024 - 2026)

- Leading technical solution architecture and implementation for a cloud-native security CTEM service utilizing the Zscaler data fabric for high-availability and strategic customer integration that mitigates OWASP Top 10 (Web & LLM) vulnerabilities and enforces kernel-level runtime security isolation (mitigating memfd_create risks and container escapes).
- Designing and scaling core Zscaler Internet Access (ZIA) inventory services utilizing container orchestration (k8s and Docker) to ensure high throughput for global cloud services, while optimizing deployment costs to shape revenue models for emerging AI/ML offerings.
- Engineering and deploying production-grade AI workflows for real-time threat log analysis using RAG-based agents, incorporating diverse LLMs (cloud, on-prem, self-hosted) and continuous model training pipelines.
- Spearheading advanced R&D across AI and cybersecurity to define technical roadmaps, authoring multiple AI/security patents, and driving internal adoption and best practices for cutting-edge AI/ML integration and open-source contributions.
- Earned a double-promotion (2 levels up) for exceptional technical leadership and cross-functional impact.

AMAZON WEB SERVICES, INC. (AWS), Sunnyvale, CA

Nov 2019 - Nov 2024

Software Development Engineer 2 (2022 - 2024)

Software Development Engineer 1 (2019 - 2022)

- Architected and developed AI-powered cloud security services (AWS Shield, Amazon Q) using Generative AI, establishing technical thought leadership across ML and cloud security patents.

- Provided technical leadership for engineering teams, driving business planning, defining product requirements with key stakeholders, and delivering high-impact solutions for large-scale product launches.
- Optimized critical infrastructure cost-efficiency through analysis and collaborated with Product Management on revenue projections for new AWS network service offerings.
- Led end-to-end development of core AWS networking services (Private 5G, VPC, Site-to-Site VPN, Telco Network Builder) and managed hardware acceleration integrations (Outposts, EC2, Edge Computing, Bare Metal as a Service, GPUs) across physical data centers.
- Engineered high-throughput data streaming solutions for AWS Aurora, developing an advanced BinLog manipulation library to ensure low-latency replication and consistency.
- Enhanced service reliability and operational excellence by establishing automation pipelines and leading Root Cause Analysis (RCA) efforts for critical AWS services.

F5 NETWORKS, INC, San Jose, CA

Jul 2018 – Nov 2019

Software Engineer 2 (2019 - 2019)

Software Engineer 1 (2018 - 2019)

- Built policy enforcement manager, service function chaining, and modular solutions for F5 BIG-IP.
- Integrated application delivery networking into CI/CD, conducted SDN controller and RFC feasibility studies.
- Collaborated cross-functionally to deliver innovative products, provided technical leadership and mentoring.

KYPHER, LLC, St. Louis, MO

May 2017 - Aug 2017

Application Developer Intern

- Full Stack Developer for Kypher™ HIPAA Compliant Messaging application.

R.O.C. NAVY, FFG-934, Taiwan

Jul 2015 - Jun 2016

Sonar Technician Seaman (Mandatory Military Service)

- Operated and maintained sonar and underwater fire control systems.

MOREMOTE, INC., Taipei, Taiwan

Jul 2014 - Jun 2015

Software Engineer

- Developed applications for two IP-Camera projects (BBCam™ and Ability HomeCam™).
- Implemented microservices with containerization, REST, Kafka with P2P technology.
- Performed strategic assessments, feasibility studies, software release maintenance.
- Worked directly with OEM clients to provide technical consultation and solution delivery.

PUBLICATION

- **A Distributed Virtual Time on Embedded System for Evaluating Cyber-Physical Systems** (ACM SIGSIM-PADS 2019) Best Paper Award 2019
- **A Distributed Embedded Linux System Smart Grid Testbed** (GCASR 2018)

PATENT

- **Self-healing and resiliency in radio-based networks using a community model.** Dec 2022
- US 11,888,701 B1, Issued Jan 2024
- **Enhancing availability of radio-based applications using multiple compute instances and virtualized network function accelerators at cloud edge** Aug 2022
- US 11,937,103 B1, Issued Mar 2024
- **Hybrid Automotive Cybersecurity Monitoring Service** Feb 2025
- US 2026/0247154 A1, Pending Aug 2026
- **Confidential (AI/ML Entropy-weighted Clustering)** Jun 2026
- Patent pending
- **Confidential (AI Evidence-Gated HA)** Jun 2026
- Patent pending
- **Confidential (AI CTEM)** Oct 2025
- Patent pending
- **Confidential (AI Threat Detection)** May 2025
- Patent pending
- **Confidential** Dec 2022
- Patent pending